

**MATH 385 Group Project
Project Report
Due 12/14/24 11:59 pm**

Group 4

Group Members:

Minh Le

Gabe Hackman

Nicholas Harrell

Matt McDonald



**MATH 385 - Introduction to Mathematical Modeling
Professor - Hye-Won Kang
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Introduction

In the era of data-driven sports analytics, where teams work day and night to gain a competitive edge, understanding the factors influencing player performance is vital to a team's success. While this area has been extensively researched, one aspect often overlooked is the role of environmental factors such as temperature, altitude, weather, and airflow (quantified by Reynold's number) in shaping athletic performance. The group chose the NFL as our focus because its national scope across varying environments makes it a prototypical model for studying the impact of environmental factors on athletic performance, with potential applications to other sports and athletic disciplines.

The team's study specifically targets quarterbacks and wide receivers, two positions we hypothesize to be mainly influenced by environmental conditions. Narrowing the group's focus to these positions simplified data processing, reduced dimensionality, and sharpened the analysis. This approach serves as a starting point for future research, which could extend to additional positions, teams, and sports.

The primary objectives of this project are:

1. How does the temperature affect quarterback performance overall?
2. How does altitude difference affect wide receiver performance?
3. How do multiple environmental factors collectively impact performance?
4. How do height and weight influence quarterback and wide receiver performance?
5. Which regression model best fits the relationship between performance and environmental factors?

The group hypothesized that the data would form a bell curve or concave quadratic curve. This means that there would be some optimal weather conditions for peak performance. Regarding temperature, players were expected to perform worse in freezing or furnace-like temperatures. For altitude difference, players were expected to perform better at home (0m altitude difference). Lower wind speeds were hypothesized to correlate with better performance. Similarly, Reynold's number, a function of wind speed, temperature, and altitude, was expected to follow a similar trend. Regarding player height and weight, the group hypothesized a bell-curve relationship. Players who are too small or too large might face disadvantages in strength, speed, or agility.

Methods

Given the extensive data in a single NFL season, the team limited our analysis to the 2023 NFL season. The group collected data from multiple sources:

1. Pro Football Reference: This database provides detailed player statistics, including fantasy points, offense percentage, position, and other metrics.
2. World Stadium Database: This database provides information on stadium altitude and location.
3. NFL Game Schedule: This dataset provided information on game dates and times.
4. Meteostat API: This API provided historical weather data, including temperature, humidity, wind speed, precipitation, and atmospheric pressure.

To prepare the data for analysis, the team merged datasets from Pro Football Reference, World Stadium Database, NFL Game Schedule, and the Meteostat API into a single data frame using Python and Jupyter notebooks (.ipynb). The group used libraries such as Pandas and Numpy for data manipulation, matplotlib for visualization, and Sklearn and scipy for statistical analysis and modeling. Additionally, we removed players with zero offensive percentages in a game and trimmed the dataset by 10% to minimize the influence of outliers.

Before developing the group's models, the group established a couple of key assumptions. First, the team assumed that all stadiums have open roofs and uniform field conditions and that HVAC systems contribute negligible heat exchange, allowing us to focus solely on external environmental factors. Fantasy points, a commonly used metric in sports betting to quantify player performance based on statistics such as touchdowns, yardage, and receptions, was equated to a direct measure of player performance, avoiding the complexity of creating or implementing a separate player performance model. Additionally, the group assumed that weather conditions obtained through the Meteostat API at halftime represented the entire game and that games labeled with the same weather conditions shared the same environmental impacts. The team assumed that weather conditions directly or indirectly influence field temperature, subsequently impacting player performance. Lastly, the group presumed that the weekly game schedule ensures comparable recovery and preparation intervals for players and that all data used in the analysis is accurate.

Models

The group used multiple forms of regression to fit curves to our data, including linear, quadratic, exponential, logistic, and power regressions. The formulas are as follows:

$$\begin{aligned}\text{Linear regression: } y &= ax + y(0) \\ \text{Quadratic regression: } y &= \pm ax^2 \pm bx \pm c \\ \text{Exponential regression: } y &= ae^x \\ \text{Logistic regression: } y &= \ln|a| + b \cdot \ln|x| \\ \text{Power regression: } y &= [ax \pm x(0)]^{0.5}\end{aligned}$$

The group created scatter plots with environmental factors (temperature, altitude, wind speed, Reynold's number) on the x-axis and fantasy points per percent offense on the y-axis. To better understand the trends within this data, the group added interpolated lines to the scatter plots to smooth out the data. Additionally, heatmaps were used to visualize the density of data points, particularly in areas with significant overlap. The team also explored the potential impact of player height and weight on performance. However, these factors were not the primary focus of our project.

Following Dr. Kang's recommendation, the group created a couple of multivariable models to analyze the effects of temperature, weather conditions, altitude, and wind speed on player performance. This model accounts for interactions between variables, potentially providing more insight than the group's single-variable approaches. Notably, the weather condition data obtained from the Meteostat API was categorical (e.g., 1 = clear, 2 = cloudy, 3 = light rain), which is unsuitable for modeling. Instead, the team used a grading system. Each condition was assigned a performance impact score between 10 and 1, ranging from optimal (benefiting performance) to poor (detrimental to performance). The formula is given below:

$$y = f(x_1, x_2, x_3, \dots) + \varepsilon$$

To verify and reinforce these findings, the group applied a Uniform Manifold Approximation and Projection (UMAP) to the same features as our multivariable models. This is a dimensionality reduction method that preserves data relationships and structure. This allowed us to visualize the possible relationships between multiple environmental factors and player performance.

The final method of multiple weather condition analysis used Reynold's Number (Re); the formula is below.

$$Re = VL_c/\nu = V\rho L_c/\mu;$$

V = Wind velocity,
 L_c = characteristic length,
 ρ = density,
 ν = kinematic viscosity,
 μ = dynamic viscosity.

The idea behind using Re was that it is a function of wind speed, temperature, and altitude (air density is inversely proportional to altitude, i.e., the higher you go, the less air there is). Re is also an indicator of fluid flow characteristics, being that it will allow the mathematician to calculate into a non-dimensional number if the flow is laminar, transient, or turbulent based on Re value.

$$Re \leq 5 \cdot 10^5 \rightarrow \text{laminar}, 5 \cdot 10^5 < Re \leq 10^8 \rightarrow \text{transient}, Re > 10^8 \rightarrow \text{turbulent}$$

Result

The results from the various graphs and modeling done by the group have led to the conclusion that weather conditions and physical player attributes do not correlate with player performance as a whole. All the graphs were filled with noise, meaning that no matter what form of regression or data analysis the group applied to the data, the graphs simply did not give valuable/useful information. We initially compared player performance to when they played in the season, which is shown in Figure 1. This graph shows no strong correlation between player performance and when they play, which is the first indicator that our hypothesis may be flawed. As the fall season progresses, the average temperatures of games gradually decrease, which would impact player performance. Because said drop in performance does not manifest itself in our plot, we likely have a hypothesis that can not be verified by our data.

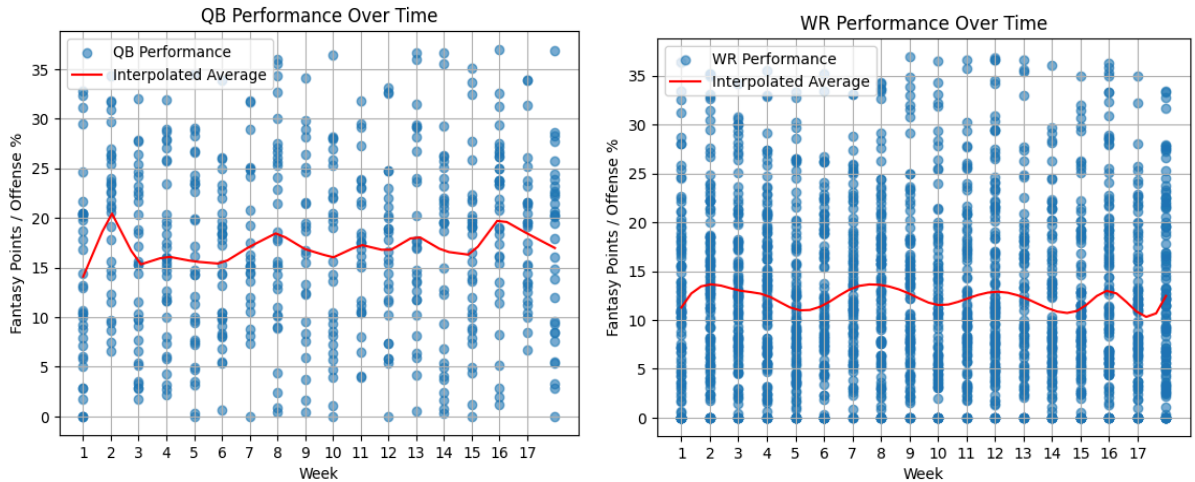


Figure 1: Average Performance of Players over a Season

Disregarding our assumption that temperature decreases as the season progresses, we reach our next analysis, which involves comparing temperature directly to player performance. This relates back to our hypothesis that temperature has a direct effect on player performance. If our hypothesis were correct, we would expect to see a quadratic relationship between temperature and performance. However, reviewing Figure 2, it becomes apparent that there is again no strong relationship between temperature and player performance. The interpolated average of performance resembles that of a random process more than a direct relationship.

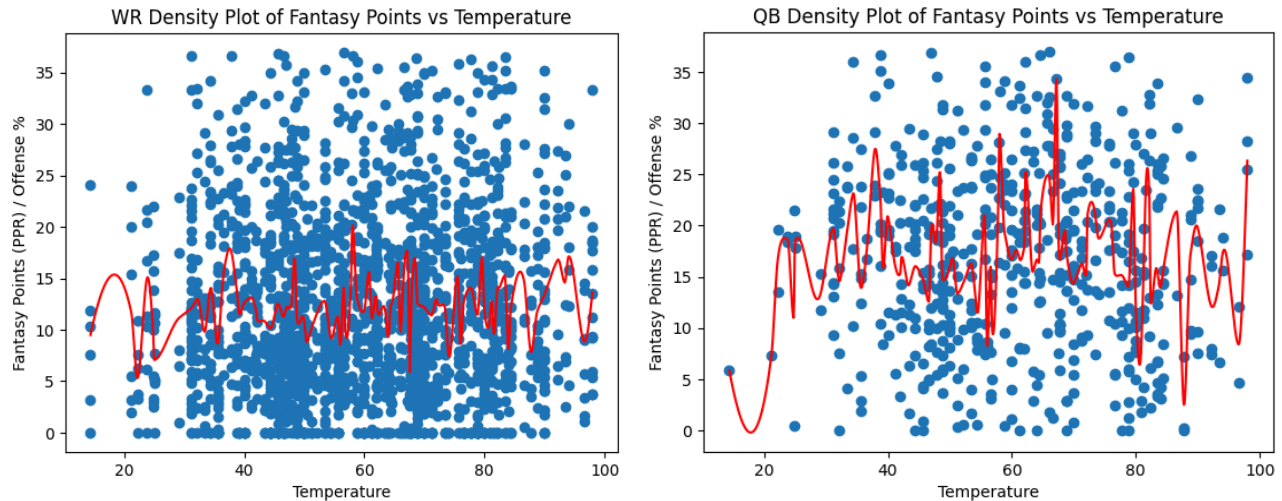


Figure 2: Average Performance of Players vs. Temperature

Since our hypothesis was not supported by the data we found, we instead planned to investigate why this was happening. Our first alternative hypothesis was that instead of playing temperature directly affecting performance, the difference in temperature between a team's practice temperature and game temperature would have an impact. This belief is based on the idea that players would become more adapted to the temperature they spend the most time in (when they practice and play at home), and a different temperature would come as a surprise to the body, leading to a change in performance. Figure 3 illustrates this information, but again, it becomes hard to find a direct relation between temperature difference and player performance.

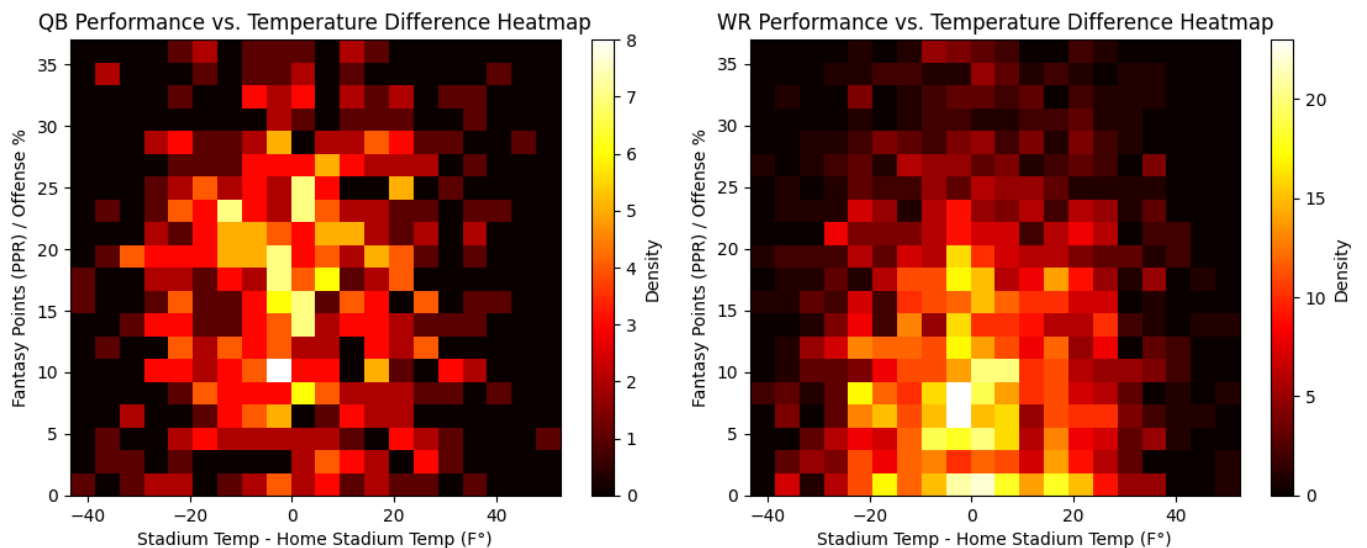


Figure 3: 2D Histogram of Performance vs. Temperature Difference

A heatmap is used to illustrate performance vs temperature difference to show how common a performance level is for a given temperature difference. For instance, at a 0-degree difference, players were more likely to earn a higher score than a low one. However, the heatmap could be misleading, as there is a lot less data at the extremities of the x-axis, and not many games are played at this temperature. To counteract this fact, a normalization of the data should have occurred for each bin. Since there still is

not a strong relation between the data, the last thing we wanted to look at was the difference in altitude. This shares a similar idea to that of the difference in temperatures, but instead of adapting to the temperature of an environment, the player would adapt to the amount of oxygen in the air. This idea is based on the fact that as altitude increases, pressure and the abundance of oxygen in the air decreases. Our findings are displayed in Figure 4. A similar problem takes place in this 2D histogram, as there is a lot less data available for games played at a high altitude difference.

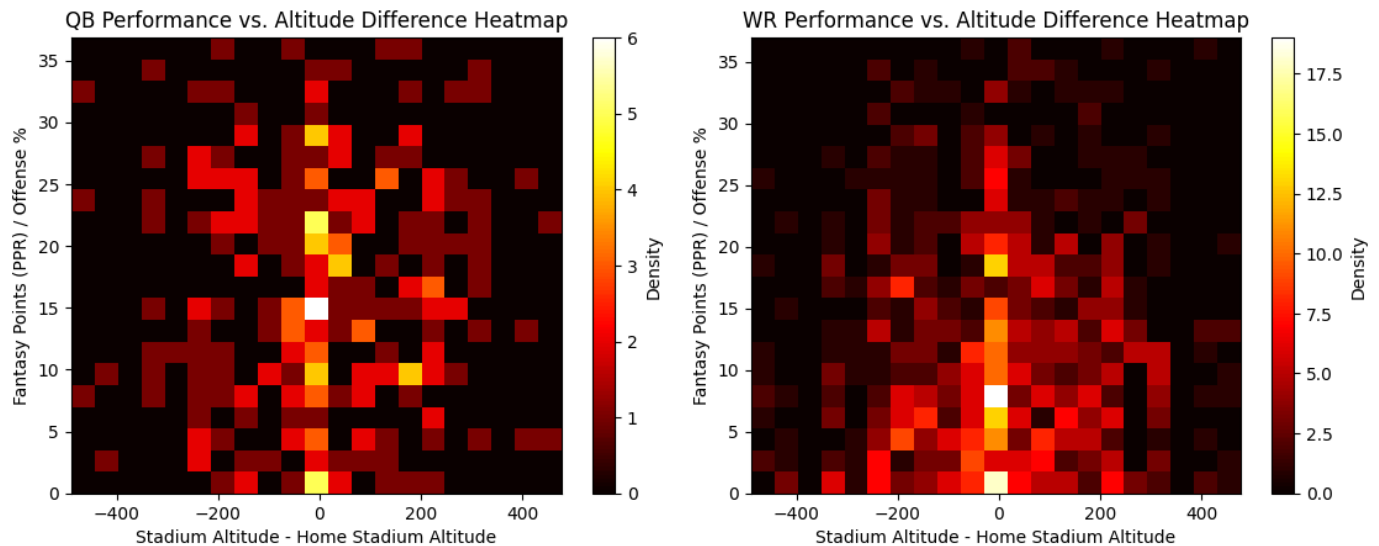


Figure 4: 2D Histogram of Performance vs. Altitude Difference

Overall, our data shows little correlation between player performance and environmental factors, so it is hard to create a model that would accurately predict or extrapolate performance for a given temperature. Despite this, we tried a few different regression models to see how closely we could approximate our data.

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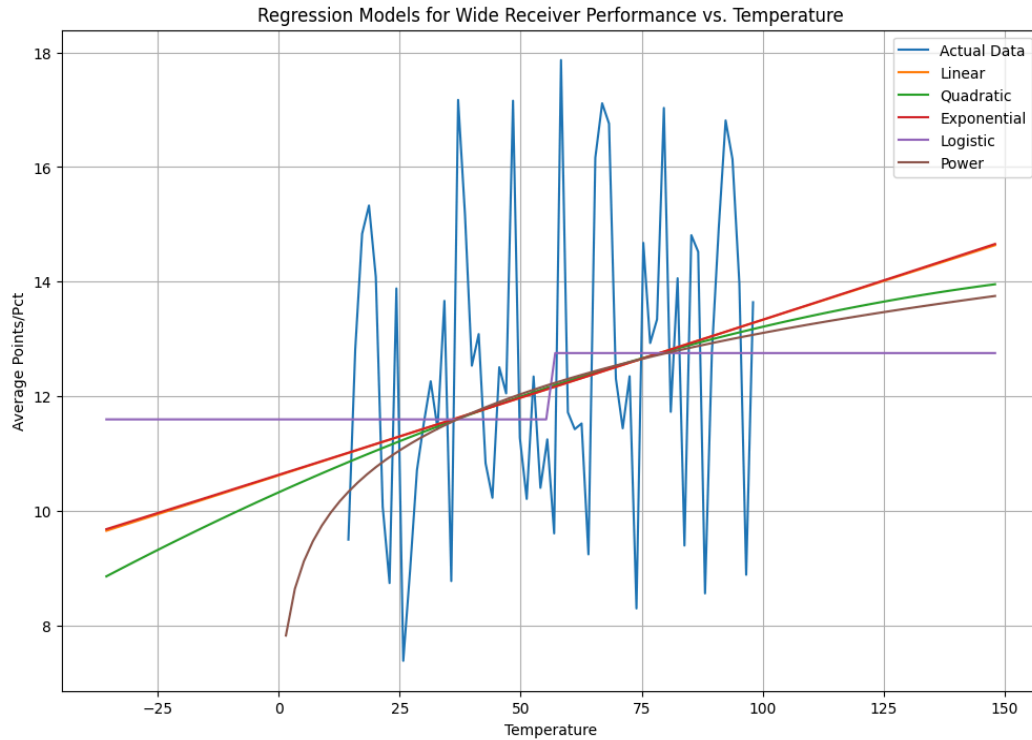


Figure 5: Regression Models for Performance vs. Temperature

As exhibited in Figure 5, no model selected can closely match the data we provided. A regression model that likely would have fit this data is Sinusoidal Regression, though its significance would be questionable as a player's performance would be bounded to some interval, implying that their performance at 72 degrees would be somewhat close to their performance at 200 degrees. The best model would likely follow some form similar to $X(t) = |a \cdot e^{-|bt|} \cdot \cos(ct)|$ or that of a Fourier Series. Regardless, the nature of the data implies that it could be a random process filled with noise, so it may be more appropriate to utilize stochastic analysis to extract meaning.

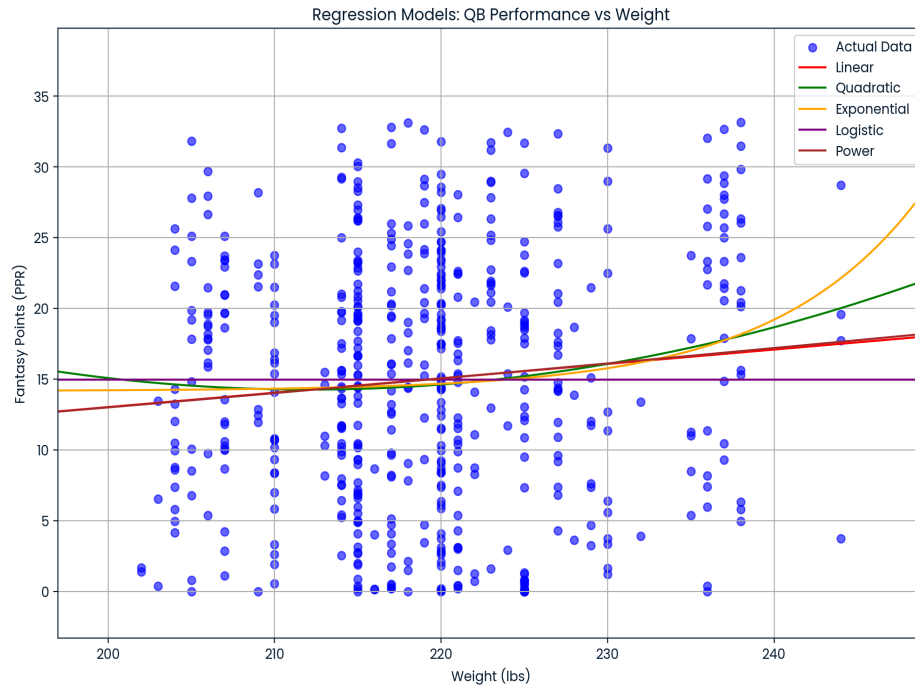


Figure 6: QB Performance vs Weight

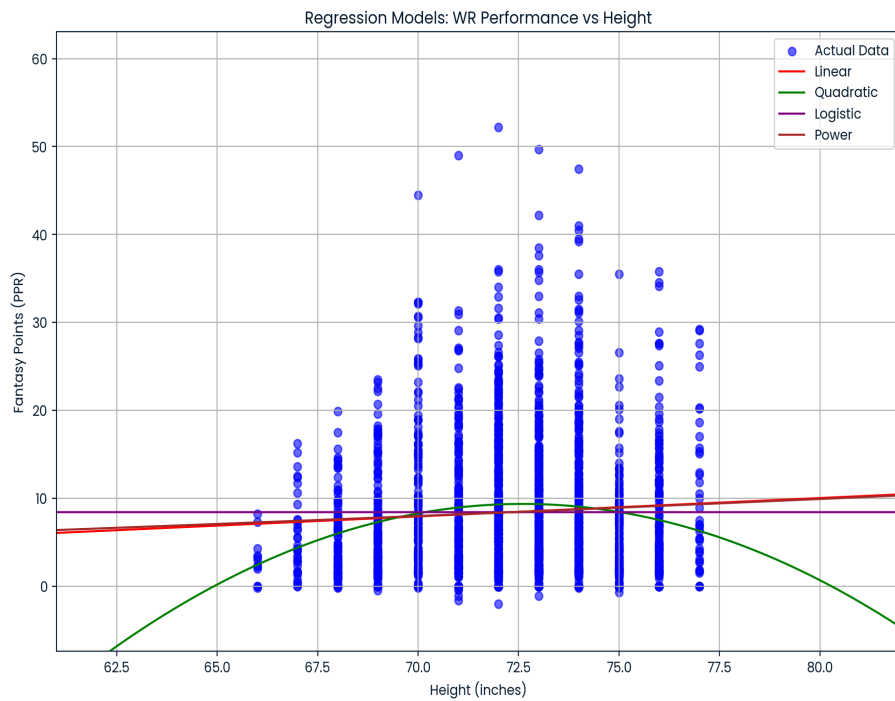


Figure 7: WR Performance vs Height

Due to these complications, the team explored additional factors, such as player height and weight for both QB and WR, as seen in Figures 6 and 7. While there is a slight trend visible in both of these figures below, they encounter the same issues as the single variate environmental factor models.

The team also explored multivariate models, including linear regression shown in Figure 8, XGBoost regression, and interaction terms, however, these models ran into the same problem as the single variate models.

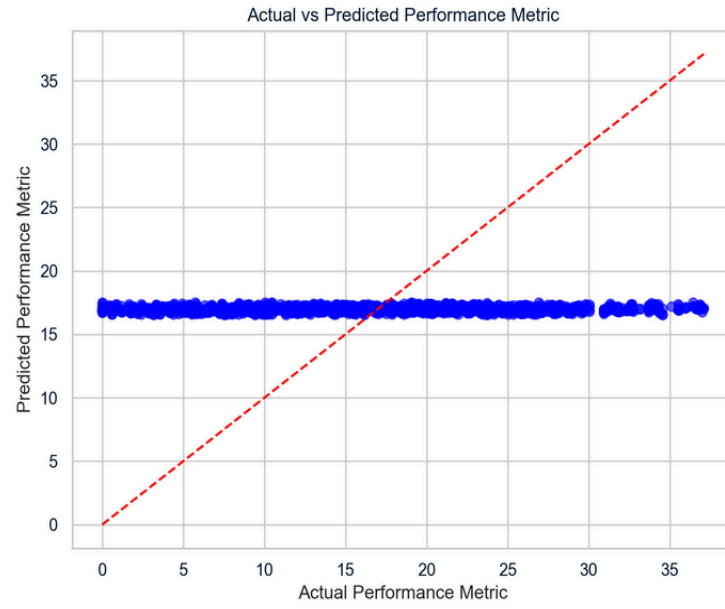


Figure 8: Multivariable Linear Model

The model produced coefficients near zero, meaning that it minimized error by predicting the average performance metric regardless of the input factors. The coefficients were as follows: temperature (-0.1825), weather (0.0485), altitude (0.0023), and wind speed (-0.0707), with an intercept of 16.9918. This resulted in a Mean Squared Error of 82.06 and an R-squared value of approximately 0.

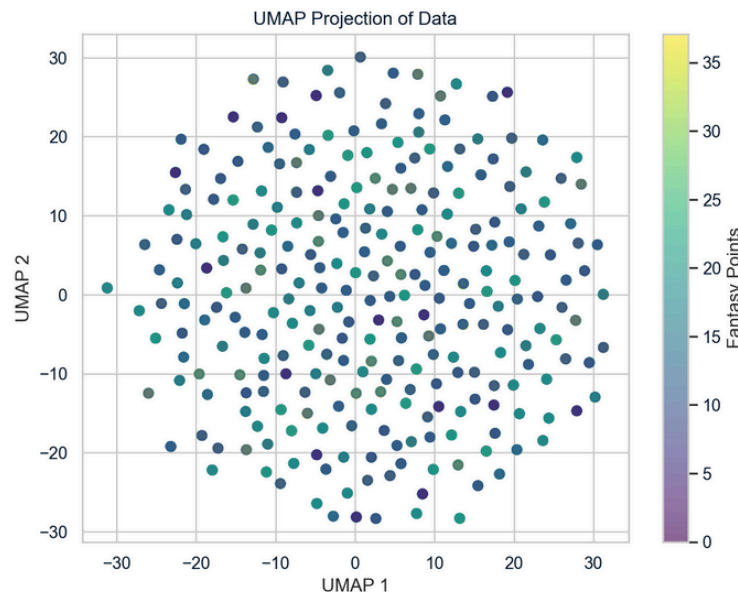


Figure 9: UMAP of QB Performance and Environmental Conditions

The UMAP analysis in Figure 9 further confirms these less than stellar results. There is no clustering or discernible pattern among these features: temperature, weather, altitude, and wind speed and

the performance metric. Therefore, the features do not have a clear relationship or have no relationship with the performance metric of players.

Discussion

The effects of environmental factors such as weather conditions, temperature, and altitude proved to be far less significant than initially hypothesized for both quarterbacks and wide receivers, showing minimal to no measurable impact on player performance. The multivariable model failed for the same reasons. Physical attributes such as weight and height also have little effect on player performance. Ultimately, the group feels unable to confidently answer the question, "Which regression model best fits the relationship between performance and environmental factors?" While the team observed a slight bell curve trend in the WR Performance vs Height plot, the trend was minor, and the other three physical attribute models showed no correlation.

One of the key assumptions the group made was treating outdoor stadiums as equivalent to domed or closed-roof stadiums, which unnecessarily oversimplified the analysis. Approximately one-third of the stadiums in our dataset fall into this category, with 10 featuring domes and 6 having retractable roofs. Excluding these stadiums and/or distinguishing between open and closed-roof states would have been a significant improvement for our models and could've helped show stronger correlations between environmental factors and player performance. Another environmental complication was the assumption that weather conditions, temperature, and wind speed, obtained at half-time, were consistent throughout an entire game. Wind speed, in particular, could be researched more thoroughly, accounting for wind direction.

Another area for further research is accounting for team matchup context. Incorporating factors such as weekly defensive rankings, team records, and head-to-head matchups, which are likely to have a high influence on player performance, may help to isolate the more subtle, yet potentially significant, effects of environmental conditions. While the team believes this analysis has the potential to be very successful, it would drastically increase the dimensionality and volume of data. Alternatively, focusing on a specific team could help build a deeper understanding of their player performance trends and compare them against other teams.

Lastly, future research could shift focus from predictive modeling to trend analysis by using a time-series approach and expanding the dataset beyond the 2023 season. This could help researchers evaluate how players adapt to environmental conditions over time. Additionally, exploring alternative metrics for player performance, rather than relying solely on fantasy points normalized by offensive percentage, could provide a more comprehensive evaluation of player output. In addition, exploring a different sport where player substituting players is less prevalent, such as long-distance running or soccer, may help reduce data inconsistencies caused by mid-game substitutions.

In conclusion, this analysis highlights the potential for deeper insights into how environmental factors impact player performance. While the current models laid a foundation, refining the approach and expanding the scope could lead to stronger correlations and trends. Therefore, by addressing these limitations, incorporating more detailed data, and exploring alternative methods, future research could offer a more accurate and comprehensive understanding of the effects of environmental conditions on performance.

References

Pro Football Reference. *Pro Football Statistic and History*. Sports Reference LLC. Retrieved 11-15-2024, from <https://www.pro-football-reference.com/>

World Stadium Database. *World Stadium Database*. Retrieved from 11-15-2024, from <https://www.worldstadiumdatabase.com/>

K. Mantzios *et al.*, “Effects of Weather Parameters on Endurance Running Performance: Discipline Specific Analysis of 1258 Races,” *Medicine & Science in Sports & Exercise*, vol. 54, no. 1, Nov. 2021, doi: <https://doi.org/10.1249/MSS.0000000000002769>.